

1. Explain the necessity of highway planning in Nepal. What are the uses of fact-finding surveys? How are these used and interpreted.

Answer:

The systematic process of developing a road network to meet present and future transportation needs efficiently and economically is known as highway planning. Highways in Nepal commonly face the problems of landslide, slope failure, river erosion, flooding, poor drainage etc. The hill roads have unstable geology and monsoon rainfall while river corridors face erosion and debris flow problems. The rugged mountainous topography, scattered settlements, and complex landscapes make it essential to carry out serious highway planning.

We can take an example of Muglin-Narayngadh highway section which is highly vulnerable to continuous landslides and falling rocks around Tuin Khola. Heavy rainfall triggers the landslide causing full and partial road blockages. Another example of Banepa-Bardibas highway can also be taken. It is constructed under the assistance of Japan which consists of one of the most remarkable drainage facilities in the section. However, the problem of subsidence and collapse of stretches are faced near Nepalthok-Khurkot, making maintenance difficult and traffic unreliable.

The necessities of highway planning in Nepal are as follows:

- i. It connects the remote hills and mountainous regions with urban centers.
- ii. It reduces the regional disparities by improving access to education, healthcare and markets.
- iii. Proper alignment and road network planning minimizes the travel distance and vehicle operating costs.
- iv. It strengthens the social, cultural and administrative links among different regions.
- v. Proper geometric design and traffic planning also helps in reducing accidents.
- vi. Highway planning also ensures optimum investment and prioritization of road projects.

The process of collection of data requires for highway planning, design, construction and future expansion is known as fact-finding surveys. There are several fact-finding surveys that can be carried out. Few are listed below with their interpretation:

- i. Traffic volume survey:
This survey helps to collect the number of vehicles passing through a particular section of a road at a specified point of time to determine the pavement thickness, road capacity etc.
- ii. Origin and destination survey:
This survey is very useful in transportation planning, design of bus stops, bus park by identifying the travel patterns and helps to select the road alignment.
- iii. Speed and delay survey:
This survey is useful in traffic planning studies, evaluation of congestion, capacity, service level and so on.
- iv. Topographical survey:
Topographical survey is helpful in fixing the economical road alignment and geometric design.
- v. Soil and geological survey:
The survey determines the subgrade conditions, slope stability as well as pavement design.

Fact-finding surveys are used:

- i. To estimate present and future traffic demand.

- ii. To identify the need for new roads or road improvements.
- iii. To determine suitable road alignment.
- iv. To estimate project cost and economic feasibility.
- v. To study population growth and land use.
- vi. To assess topography, geology, and drainage conditions.
- vii. To identify accident-prone locations.
- viii. To prioritize road development projects.
- ix. To assess environmental and social impacts.
- x. To prepare long-term transportation plans.

2. **The radius of a horizontal circular curve is 10 meters, the design speed is 50 kmph, and the design coefficient of lateral friction is 0.15. Calculate the super elevation if full lateral friction is called into play.**

Answer:

We know,

$$e + f = \frac{V^2}{127 R}$$

where:

e = super elevation

f = coefficient of lateral friction = 0.15

V = design speed = 50 km/h

R = radius of curve = 10 m

Substituting the values,

$$e + 0.15 = \frac{50^2}{127 \times 10}$$

$$e \approx 1.82 \text{ i.e. } e \approx 182\%$$

This superelevation is not practically achievable.

3. **a. Define bio-engineering. Discuss its functions and limitations. [5]**
b. Los Angeles Abrasion test [5]

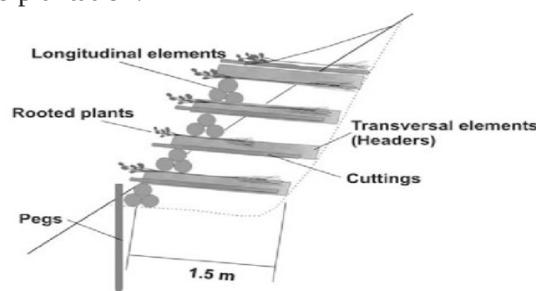
Answer:

a. Stabilization of slope using vegetative systems with small civil engineering structures is known as bio-engineering. It combines **biological measures (plants)** with **civil engineering works** to improve slope stability in an environmentally friendly and cost-effective manner.

Application of Bio-engineering helps to stabilize slopes, control erosion, and reduce landslide risks. It is widely used on hilly roads in Nepal because it is economical, environment-friendly and sustainable.

Example:

- i. Toe wall with bamboo plantation:



- ii. Jute net with grass seedling:



Principle of bio-engineering:

- The main concept of bio-engineering is the interaction between vegetative systems and small civil engineering structure.
- Plant roots bind soil particles and increase shear strength.
- Vegetation reduces the impact of rainfall and surface runoff.
- Plants absorb soil moisture, lowering pore water pressure.
- Natural cover protects slopes from erosion and weathering.

Functions of bio-engineering in hill roads:

- i. It stabilizes cut and fill slopes by reinforcing the soil with plant roots.
- ii. It reduces soil erosion caused by rainfall and surface runoff.
- iii. Bio-engineering also prevents shallow landslides and slope failures.
- iv. It protects road embankments, drainage channels, and retaining structures.
- v. It also improves slope drainage by increasing infiltration and reducing runoff velocity.
- vi. Bio-engineering also restores vegetation cover and enhances the surrounding environment.

Limitations of bio-engineering:

- i. Plants require months or years to develop strong root systems, so protection is not immediate.
- ii. Plant growth depends on rainfall, temperature, and soil conditions. Drought or frost can reduce effectiveness.
- iii. Vegetation alone cannot resist heavy earth pressures or support large structures.
- iv. Bio-engineering techniques often need more land than conventional engineering solutions.
- v. Rocky, highly erodible, or waterlogged soils may not support adequate vegetation growth.

b. Los Abrasion test:

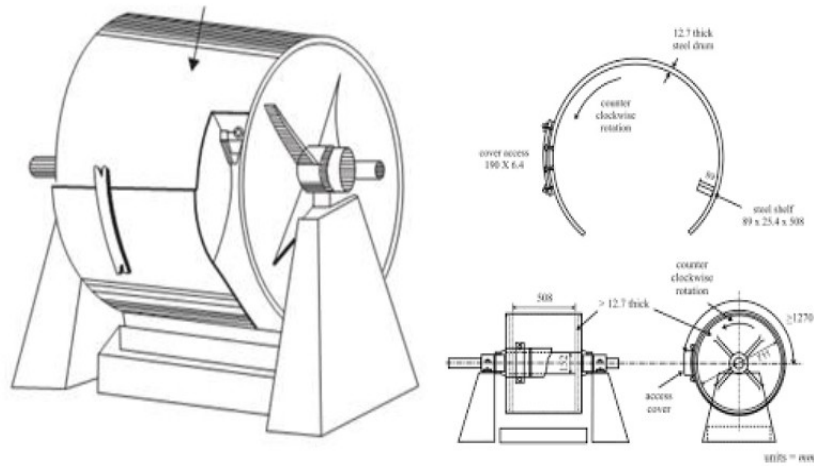
The principle of Los Angeles Abrasion Test is to produce the abrasive action by the use of standard steel balls when mixed with aggregates and rotated in the drum for specific number of revolutions and cause impact on the aggregates.

Objectives:

- i. To determine Los Angeles Abrasion Value
- ii. To find out sustainability of aggregates for its use in road construction.

Apparatus required:

- i. Machine with a hollow cylinder closed at both ends with its diameter 700 mm and length 500 mm mounted on to rotate horizontal axis.
- ii. Abrasive charge consisting of iron spheres of 48 mm diameter with weight each of 390 gm-445 gm.



The number of spheres to be used as abrasive charge and their total weight have been specified based on grading of the aggregate sample.

Procedure:

- Test sample consist of clean aggregates dried at 105°C to 110°C.
- The specified weight of the sample between 5 to 10 kg is placed inside the machine along with the abrasive charges. (W_1)
- The machine is rotated at the speed of 30 to 33 rpm for specified number of revolutions.
- The abraded aggregate is then sieved through 1.7 mm IS sieve, and the weight of powdered aggregate W_2 passing this sieve is found.
- The result of the abrasion test is expressed as the percentage wear or the percentage passing 1.7 mm sieve expressed in terms of the original weight of sample.

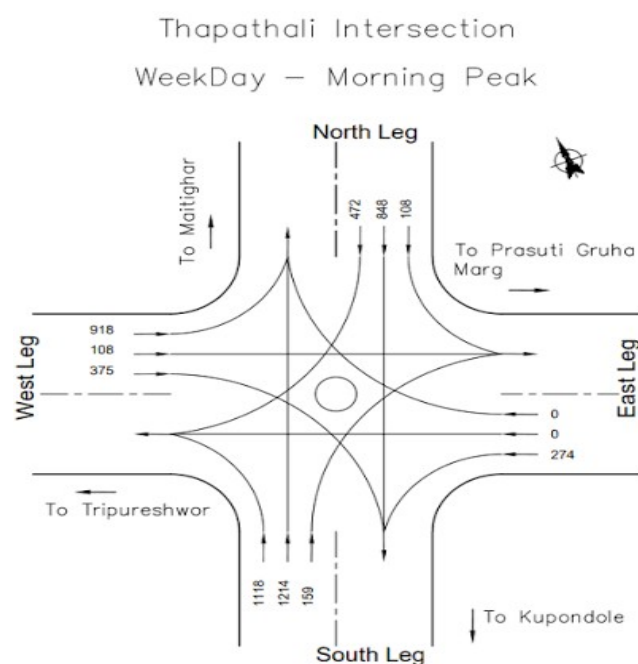
4. **Draw the highway intersection in detail and show traffic directions. [10]**

- Four-legged intersections
- Channelized intersection

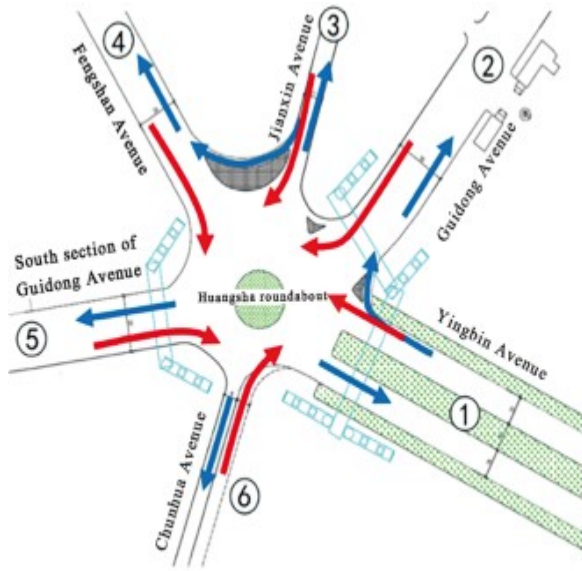
- Multi legged intersections
- Three legged intersections

Answer:

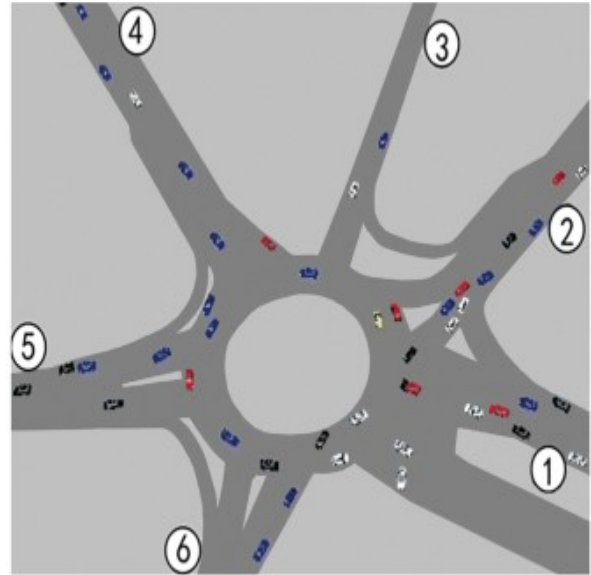
a.



b.

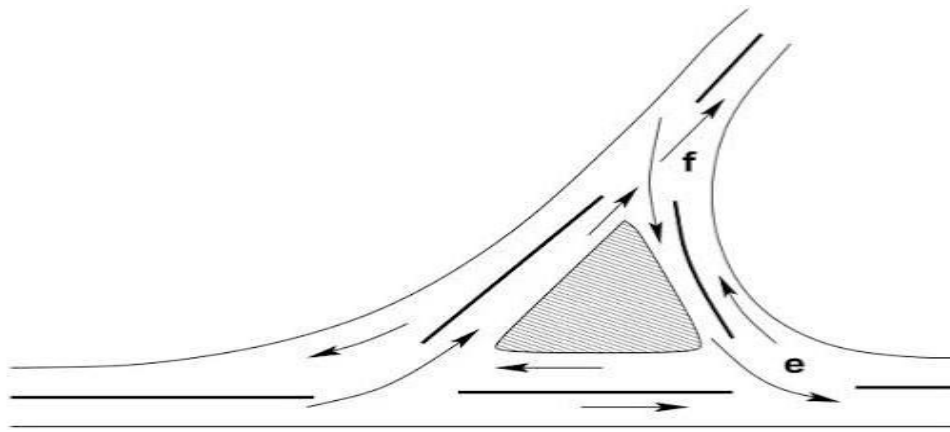


(a)



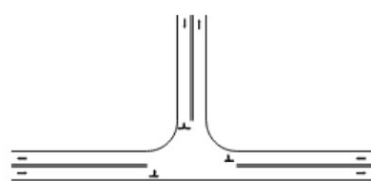
(b)

c.

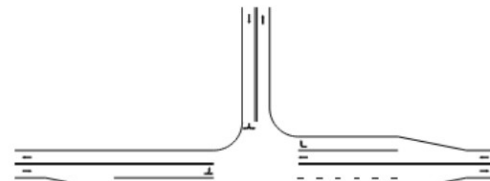


Channelization for Y Intersections

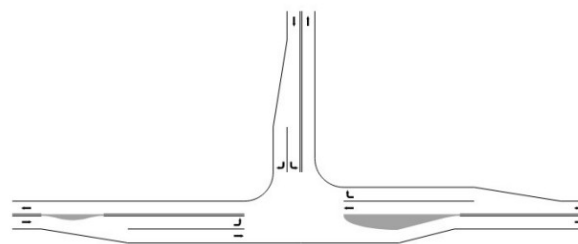
d.



Single Lane Approaches
- A -



Right-Turn Lane and Bypass Lane
- B -



Designated Lanes for Each Movement
- C -

5. What do you understand by Pre-tensioning and Post tensioning in prestressed concrete bridges? Explain the special features of prestressed concrete.

Answer:

Prestressing is one of the most important methods for preventing failure in concrete bridges. Plain concrete is weak in tension which easily forms cracks. These cracks lead to corrosion of reinforcement ultimately leading to the failure of bridge section. So, to overcome this type of failure, prestressing is introduced in the concrete where concrete is subjected to compression *before* service loads are applied, so that when bending loads are subjected, they have to overcome that compression before any tension or cracking develops.

There are two methods of prestressing:

- i. **Pre-tensioning:** usually done off-site in a precast plant maintaining very high-quality control, since strands are tensioned and concrete is poured in controlled factory conditions. It's used for standard precast girders, deck panels, and beams.

Advantages:

- a. The method is economical for mass production.
- b. It has good bond between concrete and steel.
- c. High-quality factory-controlled production.

Dis-advantages:

- a. Pre-tensioning requires heavy end anchorage and casting beds.
- b. It is not suitable for very long-span members.
- c. Transportation of large precast members may be difficult.

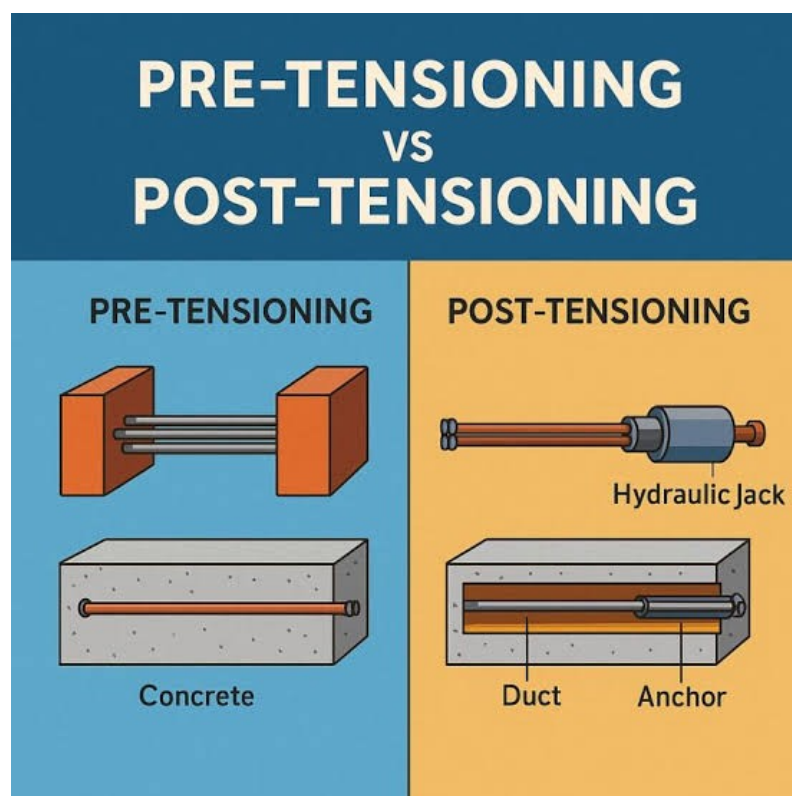
- ii. **Post-tensioning:** usually done on-site, especially for long-span bridges, box girders, and segmental construction. It is the method in which concrete is cast first, and after it gains the required strength, the steel tendons are tensioned using hydraulic jacks.

Advantages:

- a. It is suitable for long-span structures.
- b. It can be carried out at the construction site.
- c. It is better continuous structures.

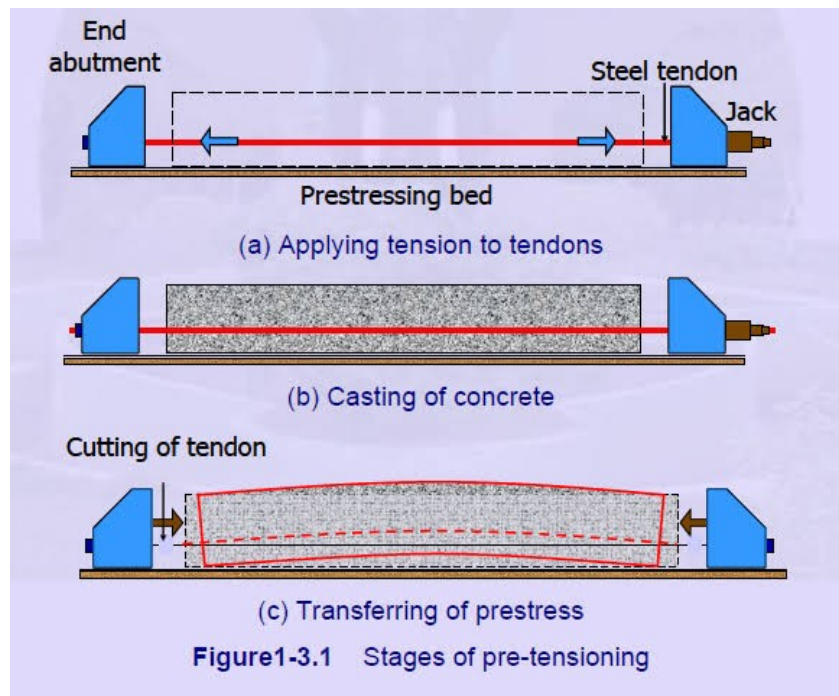
Disadvantages:

- a. Post-tensioning is more expensive.
- b. It requires skilled manpower and skilled equipment.
- c. It has greater maintenance and quality-control requirements.



Prestressed concrete:

- Prestressed concrete is concrete in which internal compressive stress is intentionally introduced before or during service so it can better resist tensile stresses from loads.
- It is widely used for long-span, slender, and crack-resistant structures such as bridges, beams, floors etc.
- There are various methods by which prestressing can be imparted in concrete. They are:
 - a. By using flat jacks.
 - b. By circumferential wire winding for developing hoop compression.
 - c. By use of longitudinally tensioned steel embedded in concrete.
 - d. By displacement and rotation of one part relative to another.
 - e. By use of expansive cement.



- Special features of prestressed concrete are listed below:
 - i. Prestressing minimizes the tensile cracks in concrete.
 - ii. It allows thinner and lighter beams, reducing the dead load of the structures.
 - iii. It also reduces the water seeping inside the cracks which thus prevents corrosion of steel.
 - iv. Prestressed concrete performs well under repeated and dynamic traffic loading.
 - v. It also reduces number of piers to be used and foundation costs.

6. What are the different types of soil classification? Describe the soil classification by unified soil classification system method.

Answer:

The arrangement of soils into different groups such that the soils in the same group shows similar behavior is known as soil classification. There are seven types of soil classification described below:

- i. MIT soil classification system
- ii. Textural soil classification system
- iii. Unified soil classification system
- iv. Indian soil classification system
- v. British soil classification system
- vi. ASTM soil classification system
- vii. ASHTO soil classification system

Unified soil classification system:

USCS was first introduced by Casagrande's.

It uses both the particle sizes and plasticity characteristics, suitable for both coarse-grained and fine-grained and fine-grained soil.

It utilizes both primary and secondary symbols.

Procedure:

i. Coarse-grained and fine-grained soils:

Coarse-grained soil:

- Soil is classified as coarse-grained soil if more than 50% retained on the 0.075 mm (No. 200) sieve.
- Divided into:
 - ✓ Gravel (G): More than 50% of coarse fraction retained on 4.75 mm (No. 4) sieve.
 - ✓ Sand (S): More than 50% of coarse fraction passes the 4.75 mm sieve.

Fine-grained soil:

- Soil is classified as fine-grained soil if more than 50% passes the 0.075 mm (No. 200) sieve.
- It is further classified using Atterberg limits into silts and clays.

ii. Classification of symbols:

For coarse grained soil:

Symbol	Description
GW	Well-graded gravel
GP	Poorly graded gravel
GM	Silty gravel
GC	Clayey gravel
SW	Well-graded sand
SP	Poorly graded sand
SM	Silty sand
SC	Clayey sand

For fine grained soil:

Based on **Liquid Limit (LL)** and **Plasticity Index (PI)**.

Low Plasticity ($LL < 50$)

Symbol	Description
ML	Inorganic silt
CL	Inorganic clay
OL	Organic silt/clay

High Plasticity ($LL \geq 50$)

Symbol	Description
MH	Elastic (high-plasticity) silt

Symbol	Description
CH	High-plasticity clay
OH	Organic clay/silt

For highly organic soils:

Symbol	Description
PT	Peat (highly organic soil)

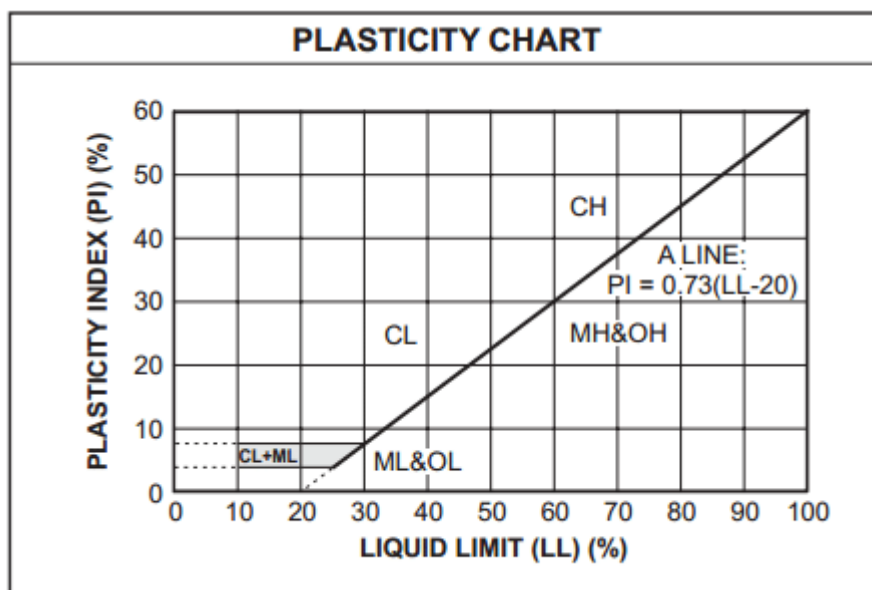
iii. Plasticity chart/A-line chart:

The plasticity chart separates silts and clays using the A-line:

$$PI = 0.73 (I_p - 20)$$

Above the A-line: Clay (C)

Below the A-line: Silt (M)



7. **Design a combined footing for two columns 32 cm x 32 cm carrying load of 60,000 kg and 40 cm carrying load of 80,000 kg. The column is spaced at 3.4 m centers and the bearing capacity of soil is 15 tonnes/m². Calculate the depth of footing, reinforcement required at transverse sections and cantilevers.**

Answer:

Given,

Column 1 = 32 cm x 32 cm, Load = 60,000 kg = 60 t

Column 2 = 40 cm x 40 cm, Load = 80,000 kg = 80 t

Centre-to-center spacing = 3.4 m

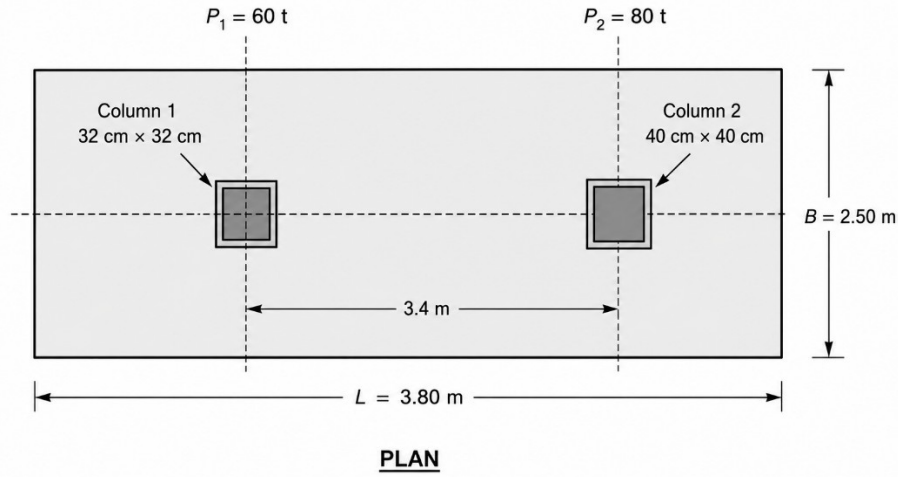
Safe bearing capacity (SBC) = 15 t/m²

Assume:

Concrete = M20

Steel = Fe415

Unit weight of footing + backfill = 10% of column load



Total load:

$$P = 60 + 80 = 140 \text{ t}$$

Including 10% for footing weight ,

$$P_t = 1.10 \times 140 = 154$$

$$A = \frac{154}{15} = 10.27 \text{ m}^2$$

$$\text{Adopt , } A = 10.5 \text{ m}^2$$

Take lighter column as origin.

$$x = \frac{60(0) + 80(3.4)}{140} = \frac{272}{140} = 1.94 \text{ m}$$

Resultant lies 1.94 m from the first column.

Assume equal projection beyond end columns.

$$\text{Let projection} = a$$

$$L = 3.4 + 2a$$

$$\text{Centroid of footing: } \frac{L}{2} = a + 1.94$$

Substituting ,

$$\frac{3.4 + 2a}{2} = a + 1.94$$

$$1.7 + a = a + 1.94$$

$$\text{Difference: } 1.94 - 1.70 = 0.24 \text{ m}$$

Hence provide projection:

$$\text{Left projection} = 0.46 \text{ m}$$

Right projection = 0.94 m

Total length: $L = 3.4 + 0.46 + 0.94 = 4.80 \text{ m}$

Width of footing: $B = \frac{10.5}{4.8} = 2.19 \text{ m}$

Adopt: $B = 2.2 \text{ m}$

Now, Soil pressure, $q = \frac{154}{4.8 \times 2.2} = 14.58 \text{ t/m}^2$, which is less than Safe Bearing Capacity (15 t/m²), Hence safe.

Using one-way shear around the heavier column,

Effective depth obtained $\approx d \approx 520 \text{ mm}$

Overall depth: $D = d + 75$

$D = 520 + 75 = 595 \text{ mm}$

Provide , $D = 600 \text{ mm}$

Longitudinal reinforcement:

Maximum bending moment near heavier column gives

Required steel , $A_{st} \approx 2100 \text{ mm}^2$

Provide , $20 \phi 150 \text{ mm c/c}$

Transverse reinforcement:

Required steel , $A_{st} \approx 1200 \text{ mm}^2$

Provide , $16 \phi 180 \text{ mm c/c}$

8. What are the different types of foundation for bridge? Describe briefly when each of these types is used.

Answer:

In bridge construction, foundations are structural elements designed to receive loads from the superstructure and transfer them safely to stable soil or rock layers. These foundations are broadly classified into:

Shallow Foundations and Deep Foundations.

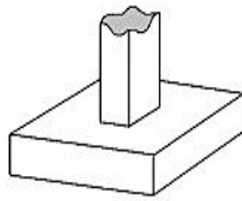
The major bridges of Nepal generally use open foundations, pile foundations, and well foundations. Among these, well foundations and pile foundations are the most common for large river bridges because they are suitable for deep alluvial soil and scour conditions. On hard rock, open or spread foundations may be used.

For example, the iconic Narayani River Cable-stayed bridge in Narayanghat-Gaindakot utilizes the pile foundation, Sunkoshi Bridge in Dolalghat utilizes the well foundation and so on.

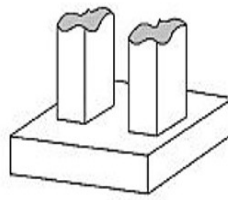
i. Shallow foundations:

- These are used when firm soil or sound rock with adequate bearing capacity is available near the ground surface.
- They are typically defined as foundations where the depth is less than or equal to the width.
- Suitable for smaller bridges, residential-scale crossings, or bridges in hilly regions where foundations can be placed directly on sound rock.

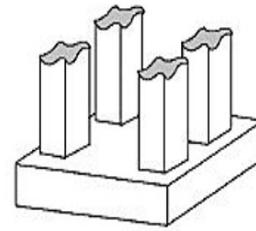
- Used when the soil with adequate bearing capacity is available near the ground surface.



Isolated Footing



Continuous footing



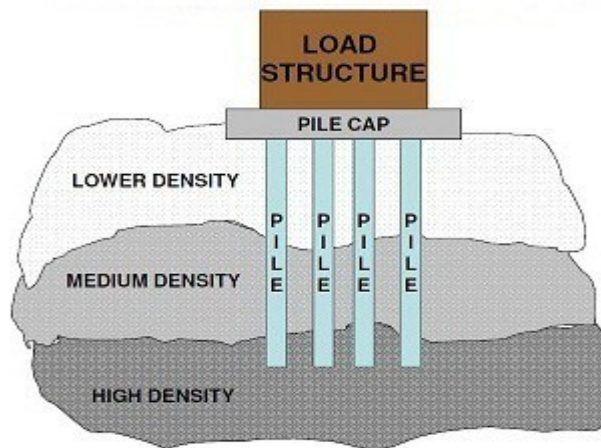
Mat footing

ii. Deep foundations:

Deep foundations are required when the upper soil layers are weak, compressible, or susceptible to erosion (scour), necessitating the transfer of loads to stronger strata at greater depths.

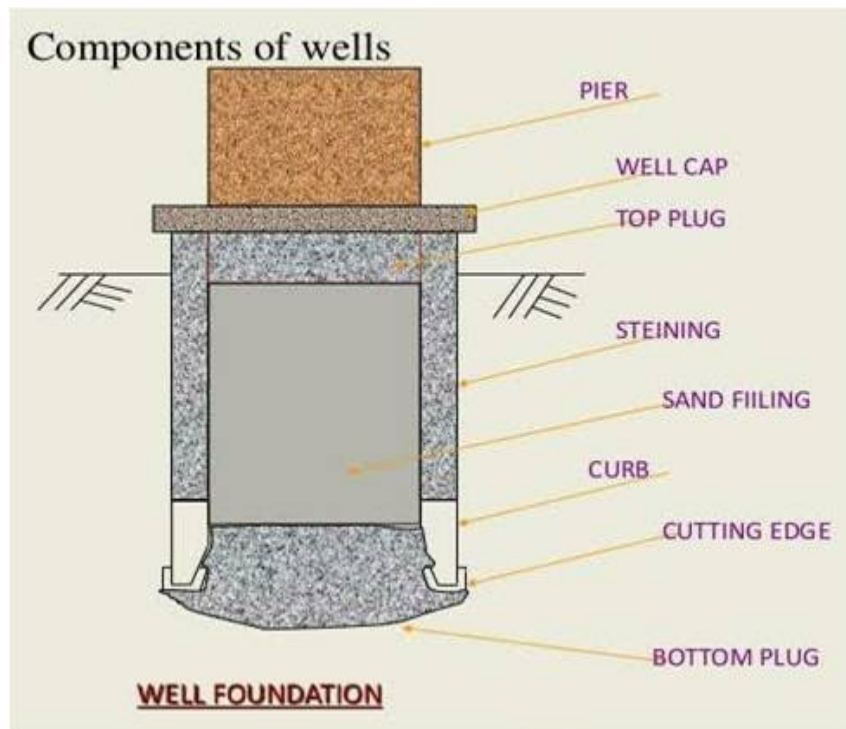
• Pile Foundations:

- ✓ Consists of long, slender members that are either driven or bored into the ground.
- ✓ They transmit loads to deeper soil strata through end bearing (resting on a hard layer), skin friction (adhesion along the pile sides), or both.
- ✓ Preferred for bridges when the bearing capacity of surface soil is low or for heavy structures in waterlogged areas.



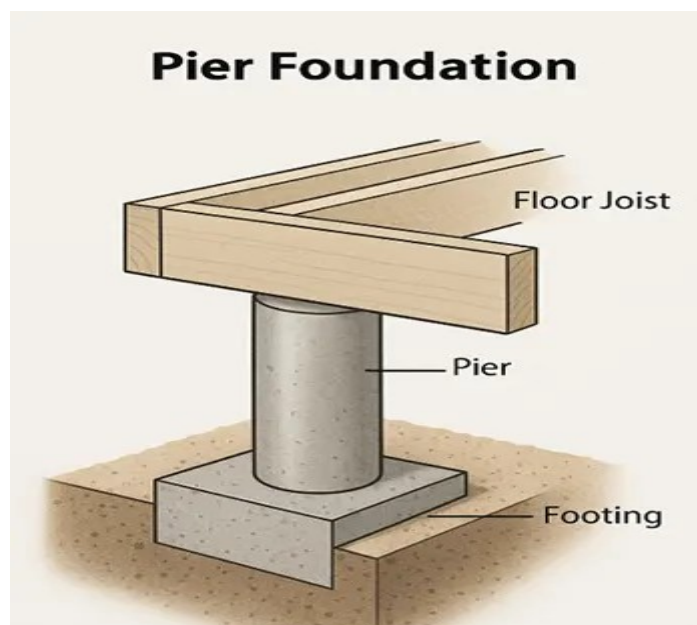
• Well Foundations (Caissons):

- ✓ These are large, watertight structures that are gradually sunk into the ground by excavating material from within them.
- ✓ Load is transferred through base resistance and side resistance after the well reaches its required depth.
- ✓ Specifically preferred for bridge piers in rivers as they provide strong resistance to heavy lateral forces and can be sunk below the scour depth to ensure stability during floods.



- **Pier Foundations:**

- ✓ These are vertical columns similar to piles but typically larger in diameter, used to transfer structural loads to a hard stratum at depth.
- ✓ Pier foundations are those foundations which transfer loads that cannot be carried by shallow foundations.
- ✓ The foundation transfers the load to the ground only by bearing.
- ✓ When the top most soil is stiff clay, it resists the driving of bearing pile and hence in that case, pier foundation is used.



The choice of foundation depends on several critical factors:

- Geological Conditions:**
 - Foundations are placed on firm, non-erodible strata whenever possible to ensure reliability.
- Scour Depth:**
 - For river bridges, the foundation must be fixed below the calculated maximum scour depth to prevent the structure from being undermined during high-velocity flood events.

iii. **Bridge Type:**

- Heavier structures like arch bridges require very strong foundations, whereas lighter steel truss or cable-supported bridges may be feasible on slightly weaker strata.